

Customer behavior Analysis for E-Commerce and Retail using Improved K-Means Clustering with Calinski Harabasz Index

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Abstract—In recent days, the customer behavior understanding is crucial for business to tailor the marketing strategies in the rapid growth of e-commerce landscapes to improve customer satisfaction. However, the traditional clustering algorithms struggled with cold start, scalability and sparsity issues which led to inaccurate results. Therefore, this research proposes an Improved K-Means Clustering with Calinski Harabasz Index (IMC-CHI) model for behavior analysis of customer in e-commerce. The K-means clustering algorithm assigns weights to the significant features to improve the clustering of new or sparse data. In addition, the CHI determines the optimal number of clusters by ensuring efficient and scalable customer behavior segmentation. Initially, the data is collected from Brazilian E-Commerce Public Dataset by Olist (BECPDO) and further preprocessed to handle missing values with K-Nearest Neighbor (KNN) imputation. Then, the recency, frequency, and monetary indicators are extracted and normalized with z-score normalization to avoid disparity. Finally, the proposed IMC-CHI is introduced to cluster the customer behaviors of e-commerce landscapes. From the results, the proposed IMC-CHI outperformed Improved K-Medoids (IKM) algorithm in terms of CH as 8100 and Sum of Squared Errors (SSE) as 34000 per 12 clusters respectively.

Keywords—calinski harabasz index, customer behavior analysis, E-commerce, improved K-means clustering, K-nearest neighbor imputation.

I. INTRODUCTION

In recent years, the customer behavior analysis has become a basis for strategic decision-making in both e-commerce and retail sectors [1]. In addition, with the increasing volume of digital transactions and the cross-channel engagement making difficult to understand how customers are interacting with platforms, products and promotions [2]. Traditionally, the behavior analysis was limited to in-store observation but the digital shift in e-commerce enabled the collection of detailed behavioral data including cart activities and purchase frequencies [3]. This enhanced data environment provides retailers and the online sellers to understand customer preference and personalize better experiences for future actions. Therefore, these developments in understanding customer behavior significantly transforming how businesses adapt the needs and

expectations in competitive marketplace [4]. Despite of data availability in e-commerce and retail, the customer behavior analysis suffers with several challenges such as the heterogeneity of customer details. In addition, the difficulties in create a unifying and interpreting the information about a customer across platforms [5]. The analysis of customer behavior becomes complex, as it is highly dynamic and context-sensitive which influence by the factors such as trends, seasons. Consequently, the businesses mostly lack with analytical expertise which required to process the behavioral data of customers to derive actionable insights from large-scale data [6]. These challenges need to be addressed with more advanced and context-aware approaches to fully unlock the value of behavioral analysis in e-commerce and retail.

These requirements motivated to use the clustering techniques to segment the similar group of customers behaviors into same clusters. Therefore, customer behavior is vital for understanding the engagement and increasing sales to maintain the long-term loyalty in landscapes of e-commerce and retail [7]. Analyzing of customer behavior enable the companies to extract the hidden patterns and identify the high-value segments to optimize product recommendations. Consequently, the integration of Artificial Intelligence (AI) and Machine Learning (ML) in behavior analysis further enables the real-time decision-making and personalized recommendations [8]. Ultimately, improving behavior analysis not only determine the competitive benefits but also ensures the customer satisfaction and trust in increasing market place [9]. The state-of-art methods for customer behavior analysis are as follows: Deep Learning (DL)-based recommendation systems which mainly focused on neural networks to learn the complex patterns but struggled with sparse and high-dimensional data [10]. In addition, the graph-based clustering methods are utilized to represent the customers and products as nodes in a graph by enabling the patterns in customer behavior but suffered with scalability issues of graph structures [11]. Moreover, the sequence-based analysis using neural networks are utilized for sequential customer behavior to capture temporal patterns but mainly lacks in handling the long-term dependencies and vanishing gradient problems [12]. The main motive of this research is to

resolve the maximum issues of existing methods with proposed IMC-CHI clustering model.

The contribution of this research is listed as follows:

- The K-Nearest Neighbor (KNN) imputation effectively handled the missing values in customer behavior data by estimating the missing values based similar interactions to preserve the data integrity and enhances the clustering accuracy.
- The Improved K-Means Clustering (IMC) utilized combined Analytic hierarchy process (AHP) and entropy method for assigning weights to ensure more accurate feature importance for effective clustering of customer behaviors.
- The Calinski-Harabasz Index (CHI) effectively validated the clustering quality to ensure the optimal cluster separation by guiding the selection of best number of clusters for resulting in more precise outcomes.

The organization of this research is structured as follows: Section 2 explains literature review, section 3 demonstrates proposed methodology, section 4 deliberates experimental results, section 5 illustrates the conclusion.

II. LITERATURE REVIEW

Pengfei Li, Chen Wang et al. [13] illustrated an RFM weighted K-means designed to enhance e-commerce accurate customer segmentation. Initially, input data recency, frequency, monetary indicators were considered and normalized them to avoid different measurement units on customer segmentation and weights were obtained with AHP-entropy method. Further, those weights were calculated using clustering to divide the customers based on similarities. Customer types were analyzed and values calculated by combining weights, select the optimal k which maximizes accuracy and used for better segmentation. However, the model struggled with low accuracy because real world data may contain noise and it consists of limited features.

Zeng yuan Wu et al. [14] illustrated a customer segmentation through an improved K-medoids clustering algorithm designed to enhance customer segmentation accuracy and efficiency. Initially, input data was considered recency, frequency, monetary and added two actions cart and favorites, refine by removing all duplicates and normalize them bring same scale to avoid different measurements for customer classification. After that each segment compute k using k-medoids and calinski-harabasz was used to determine the optimal number of clustering, further it selected the first medoid randomly then pick next one with their probability and

proportional and swap medoids with non-medoids to decrease the error rate. However, the model is struggled with accuracy and efficiency noise in the data.

Kayalvily Tabianan et al. [15] illustrated a k-means clustering approach designed to enhance customer purchase behavior on e-commerce to promote right product to the right customer. Initially, input data was considered it consists of customer purchased history. Since, the dataset contains millions of data which leads to inaccurate output results therefore, selected limited amount of data from past 3 years. The input data was preprocessed to check missing values and any other inconsistencies. K-means clustering used to divide data into similar clusters such as, views of product, product in cart, purchased where each cluster is defined with k. Further, it leads to calculate maximum result from the dataset and chooses optimal number of clusters based on criteria. However, the model struggled with overfitting because of high k values reduces the error, but raises of overfitting.

Farah Tawfiq Abdul Hussien et al. [16] illustrated e-commerce recommendation system designed to enhance customer behavior based on different applications. Initially, input data consists of the computer brands, such as Lenovo, acer and each brand assigned additional components like keyboard, mouse. After that each device is classified according to their price ranges, further used recommendation system it consists parameters used to classify products based on (like, dislike, view, purchase, rate) each product contains unique identifier to retrieve the products. It was used to build customer matrix based on their customer behavior and products in customer database. However, the model struggled with cold start because of few sales and reviews for new products that not as prior sales history or feedback.

Junhai Wang et al. [17] illustrated a decision support system for e-commerce designed to enhance customer satisfaction by providing faster and best shipping. Initially, the framework involved 3 stages firstly, assessment criteria and data it involves indicators such as speed, cost, quality and at this stage all aspects must be measured and evaluated. Next stage, data collection it was performed through wide variety of methods include surveys, interviews, observations. Effective data collections lead to provide relevant insights for further analysis. Finally, data analysis stage it processes the data which has been collected to extract patterns and trends. Further, SAW and CRITIC were effective methods used to decide weights for factors which makes decisions. However, the model is struggled with choosing the best shipping methods often it contains related factors which can affect the business sustainability.

III. METHODOLOGY

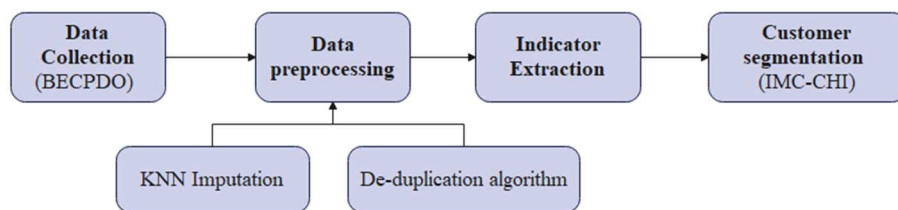


Fig. 1. Pictorial representation of proposed IMC-CHI

In this section, the customer behavior analysis has been employed with proposed IMC-CHI in e-commerce and retail.

Initially, the customer data is collected from Brazilian E-Commerce Public Dataset by Olist (BECPDO) dataset. Then,

preprocessed with KNN imputation to handle missing values and then duplicate values are removed with de-duplication algorithm. After that, the recency, frequency, and monetary indicators are extracted and normalized with z-score normalization in order to avoid disparity. Finally, the proposed IMC-CHI is introduced to cluster the customer behaviors of e-commerce landscapes. The pictorial representation of proposed IMC-CHI is given in above Figure 1.

A. Data Collection

The customer behavior analysis is done with proposed IMC-CHI model based on BECPDO dataset which is publicly available at Kaggle portal [18]. The dataset contains 100,000 orders from the different marketplaces in Brazil from the year of 2016 to 2018 respectively. Here, the dataset comprises of many features which contained price, order status, payment and the load performance for the location, attributes of product as well as the reviews. Moreover, order and the online behavior data of 37,376 customers are extracted from this dataset. The consumption time of the features is from 2017, November 18 to December 18. For clustering the e-commerce customers, 3 fields are selected including recency, frequency, and monetary indicators. The input features are further preprocessed with KNN imputation which is given in below section.

$$D_{xy} = \sqrt{\text{weight} * \text{squared distance from present coordinates}} \quad (1)$$

Where, D_{xy} represents Euclidean distance and the weight is proportion of all coordinates. The missing values handled data is further processed with de-duplication algorithm to identify and remove duplicate values from the customer data of e-commerce and retail that is given in below subsection.

2) *Duplicates Removal*: The duplicate values from the missing values handled customer data are identified and removed with de-duplication algorithm. These duplicate values are identified by comparing the records with similarity measures. For instance, the two records A and B are compared and the similarity score is computed with Jaccard similarity which is given in below Equation (2):

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (2)$$

If $J(A, B) > \theta$, where θ denotes threshold values then they are considered as duplicates and the duplicate values are removed by retaining unique values. The recency, frequency, and monetary indicator are extracted which is given in below section.

C. Indicator Extraction

The three indicators are extracted from the preprocessed data where the recency as the time interval between a customer's most recent purchase and the point of analysis. In addition, frequency demonstrates the frequency refers to the number of purchases a customer makes within a specific time period. and monetary signifies the monetary value represents the total amount spent by a customer during a specific time period. When proximity is low then frequency and value are higher consequently customer value is also higher. In addition, the standardization of the indicators avoids the impact of disparity caused by each indicator. Therefore, z-score

B. Data Preprocessing

The input customer data for e-commerce and retail is further preprocessed with KNN imputation and de-duplication algorithm. Here, the missing values are handled with KNN imputation and then the de-duplication algorithm is employed to identify and remove duplicate value that is given below section.

1) *Handling Missing Values*: The input behavioral data of e-commerce and retail customers need to be preprocessed because, it collected in a month about 100,000 pieces. Therefore, KNN imputation is employed to handle missing values and then the duplicate values are removed for effective clustering of customers. From the customers data, the information which is missing can commonly found and handled because those values have been gathered due to human error during collection or extraction processes. So, the imputer substitutes a value to handle the missing values by finding the nearest neighbors by using the Euclidean distance matrix. Therefore, this is a tool for values infilling that are lost from observations and assigns higher weights to coordinates that are non-missing and not considering missing ones in the Euclidean distance calculation and the Euclidean distance is obtained by utilizing Equation (1):

normalization is employed to normalize the data by providing mean and standard deviation. The normalized data yields the standard normal distribution and the transformation function is described in below Equation (3):

$$X^* = \frac{X - \mu}{\sigma} \quad (3)$$

Where, X^* denotes the standardized indicators, μ specifies the mean and σ signifies the standard deviation of all the samples. After performing normalization, the indicators values converted to the dimensionless data and the clustering is performed that is given in below section.

D. Customer Segmentation for E-Commerce and Retail

In this phase, the index weight for each indicator is obtained with combined entropy and AHP method. Therefore, first the AHP method constructs the judgement matrix for the indicators and calculate index weight as well as check the consistency. From the judgement matrix, eigen values and eigen vectors are calculated and eigenvector which is normalized is considered as weight vector and it is treated as index weight for w_{AR} , w_{AF} , w_{AM} respectively. Consequently, each index in judgment matrix weight is calculated and consistency check is performed. When the ratio of consistency is less than 0.1 to pass the test; otherwise, the matrix need to be reviewed and calculation formula is given in below Equations (4) and (5):

$$CI = (\lambda_{max}(A) - n) / (n - 1) \quad (4)$$

$$CR = CI / RI \quad (5)$$

Where, CI denotes the consistency index of judgement matrix and RI specifies the random consistency index, which

is selected according to the order n of the judgement matrix. Then, the entropy method assesses the amount of useful information each indicator provides. Accordingly, it assigns weights based on the informational contribution of each indicator. Here, indicator which provides greater information attains smaller entropy value and smaller entropy value indicates greater weight. The entropy method to calculate indicator weight is described as follows: Let us consider that n samples and m indicators and x_{ij} is the j -th indicator value of the i -th sample. The proportion of the i -th sample under j -th index is calculated in below Equation (6):

$$P_{ij} = x_{ij} / \sum_{i=1}^n x_{ij}, i = 1, \dots, n; j = 1, \dots, m \quad (6)$$

Then, the information entropy is calculated for the j -th index is given in below Equation (7):

$$e_j = -k \sum_{i=1}^n P_{ij} \ln(P_{ij}), i = 1, \dots, n; j = 1, \dots, m \quad (7)$$

Where, $k = 1/\ln(n) > 0$ and for j -th index, redundancy information entropy is calculated and that is given in below Equation (8):

$$d_j = 1 - e_j, j = 1, \dots, m \quad (8)$$

After calculating the redundancy, the weight for each indicator is evaluated by removing the redundancy data and the weight calculation is given in below Equation (9):

$$w_{Ej} = d_j / \sum_{j=1}^m d_j, j = 1, \dots, m \quad (9)$$

The obtained weights are specified for each indicator as w_{ER} , w_{EF} and w_{EM} respectively. Therefore, the combined weights from AHP and entropy for indicators is calculated for j -th index is given in below Equation (10):

$$w_{cj} = w_{Aj} w_{Ej} / \sum_{j=1}^m w_{Aj} w_{Ej} \quad (10)$$

Where, w_{cj} specifies the combined weight of j -th index and w_{Aj} specifies the obtained weight from AHP and w_{Ej} signifies obtained weight from entropy method. This combined weight for each indicator is further processed with CHI to find the best customer cluster. Now, the indicator weights are considered as k clusters and for each k value, the CHI is computed with below Equation (11):

$$CHI(k) = \frac{Tr(B_k)}{Tr(W_k)} \cdot \frac{n-k}{k-1} \quad (11)$$

Where, $Tr(B_k)$ specifies the between-cluster dispersion, $Tr(W_k)$ defines the within cluster dispersion and n is the total number of samples, k specifies the number of clusters. Then, the optimal cluster is selected with the highest CHI values that is given in below Equation (12):

$$k^* = \arg \max_k CHI(k) \quad (12)$$

Therefore, the highest CHI value cluster is considered as the better defined, more separated and impactful clusters which specifies most important behavior customers for e-commerce and retail. The customer levels with respect to cluster indicators are evaluated and given in below Table 1.

TABLE I. CUSTOMER LEVELS WITH RESPECT TO WEIGHTED INDICATORS

Customer categories	Recency	Frequency	Monetary	Weights	Results	Customer level
1	0.975	0.325	0.253	0.523	R↑F↑M↑	Important
2	0.928	0.053	0.032	0.356	R↑F↓M↓	General
3	0.634	0.020	0.019	0.259	R↓F↓M↓	Low-value

From Table 1, the highest weights with better indicator values according to CHI metric is considered as most important customers and other clusters are segmented as general and low-level customers. The performance of proposed IMC-CHI is evaluated and described in below section.

IV. EXPERIMENTAL RESULTS

The customer behavior analysis for e-commerce and retail is implemented with proposed IMC-CHI model. The hardware configurations intel core i3 processor and internal storage 32 GB and the processor running at a speed of 3.24 GHz. The software configurations including python programming language with scikit-learn, numPy libraries on windows 11 are considered. The clustering evaluation metrics including CH and Sum of Squared Error (SSE) are utilized and the mathematical formulae demonstrated in below Equations (13) and (14):

$$CH = (B/(k-1))/(W/(n-k)) \quad (13)$$

$$SSE = \sum (x_i - \mu_i)^2 \quad (14)$$

Where, B denotes between cluster variance, W specifies within cluster variance, k specifies the number of clusters and n signifies the total number of data points. In addition, x_i

denoted as individual data point and μ_i indicates the mean of cluster which x_i belongs to respectively.

A. Performance evaluation

The performance is evaluated for proposed IMC-CHI model with the help of various traditional models using clustering indicators. The performance models such as K-means, K-medoids and fuzzy clustering are considered. The customer importance level is compared by the proposed IMC-CHI and traditional models that is given in below Figure 2.

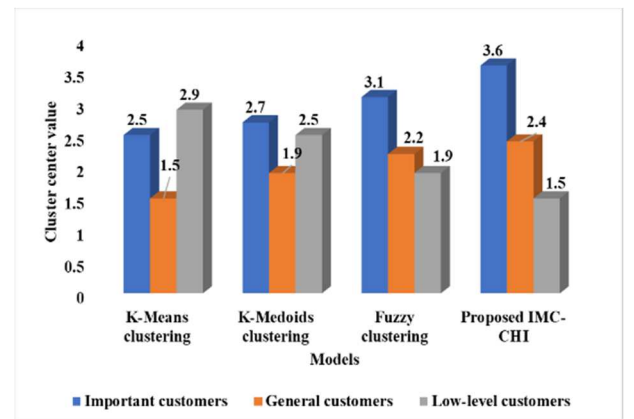


Fig. 2. Performance analysis of proposed IMC-CHI

The Figure 2 shows that the proposed IMC-CHI model performance is compared with traditional models for customer importance levels. It is clearly shown that the proposed model is having more important customer than traditional models where for 3.6 important customers are there for 3.5 customer center value respectively.

B. Comparative evaluation

The proposed IMC-CHI is compared with existing Improved K-Means (IKM) [13] clustering algorithm by using clustering evaluation metrics such as CH and SSE respectively. The comparative evaluation is given in below Table 2.

TABLE II. COMPARATIVE ANALYSIS OF PROPOSED IMC-CHI

Comparative models	CH			SSE		
	No. of Clusters					
	4	8	12	4	8	12
IKM [13]	12000	8700	7800	47000	40000	37000
Proposed IMC-CHI	13000	8900	8100	45000	37000	34000

Here, Table 2 demonstrates the comparative evaluation of proposed IMC-CHI with existing IKM [13] using CH and SSE clustering evaluation metrics. The comparative evaluation is done with respect to number of clusters. The existing IKM [13] established 7800 of CH and 37000 of SSE for 12 clusters. The proposed IMC-CHI attained better results than existing IKM [13] with ability of assigning weights and selecting optimal weights using CHI. Therefore, the proposed IMC-CHI attained better results in terms of CH as 8100 and SSE as 34000 per 12 cluster.

C. Discussion

The main objective of this research is to develop a model for customer behavior analysis in e-commerce and retail. The proposed IMC-CHI effectively overcame the drawbacks of traditional and existing approaches like k-means, k-medoids, fuzzy clustering and IKM [13] by addressing limitations that related to cluster initialization, number selection and intra-cluster compactness. The traditional k-means and k-medoids depends heavily on random initialization of cluster centroids which often leads to suboptimal convergence and poor clustering performance. The proposed IMC-CHI incorporates optimized centroid initialization strategy which improves consistency and stability of clustering outcomes. Additionally, conventional clustering algorithms generally requires manual specification of number of clusters which lead to overfitting or underfitting. The proposed IMC-CHI uses CHI which is a robust internal validation metric, to automatically determine optimal number of clusters based on compactness and separation of data points that removes dependency on heuristic or trial-and-error-based methods. Furthermore, fuzzy clustering techniques, often suffer from ambiguity in membership degrees which affects interpretability. Proposed IMC-CHI, maintains clear cluster boundaries by enhancing intra-cluster similarity and inter-cluster dissimilarity. Compared to IKM [15], the proposed model offered better scalability for large datasets, improved cluster quality and faster convergence. These improvements make IMC-CHI highly suitable for customer segmentation by purchasing pattern recognition and personalized marketing strategy formulation in e-commerce platforms. Overall, IMC-CHI provided a more intelligent, efficient and accurate approach for analyzing complex customer behavior data in real-world retail environments.

V. CONCLUSION

In this research, effective IMC-CHI model is proposed for customer behavior analysis in e-commerce by addressing the challenges of optimal cluster selection and data sparsity using the BBCEPDO. Initially, raw customer data is collected which includes various transactional and behavioral features. For ensuring the data quality and consistency, missing values are handled using KNN imputation approach. Then, RFM indicators are extracted as key behavioral attributes and normalized using z-score normalization for eliminating the feature scale bias. The improved K-Means algorithm is then applied with an optimized centroid initialization approach for ensuring robust and consistent clustering performance. For determining the optimal number of customer segments, CHI is used which ensures maximum intra-cluster cohesion and inter-cluster separation. Finally, proposed IMC-CHI model effectively clusters the customers into meaningful segments that supported targeted marketing strategies and personalized services. The proposed model significantly outperforms IKM algorithm by achieving a higher CHI score of 8100 and a lower SSE of 34000 with 12 clusters. In the future, this research will be extended by integrating temporal behavior analysis and hybrid clustering models for enhancing adaptability and scalability of customer segmentation in dynamic e-commerce environments.

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